Convolutional Neural Networks (CNNs)

Architecture, Convolutions, and Backpropagation

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Motivation: Why Convolutions?

- Images have spatial structure: nearby pixels contain related information.
- Fully Connected (Dense) Networks ignore spatial locality.
- CNNs extract local patterns (edges, textures, shapes) automatically.
- Convolutions reduce parameters while preserving spatial information.

Key Idea of CNN

Convolution Operation

A filter slides across the image and computes:

$$(\mathbf{I} * \mathbf{K})(i,j) = \sum_m \sum_n I(i+m, j+n)\, K(m,n)$$

- Filters learn visual features.
- Multiple filters produce multiple feature maps.
- Deep layers build hierarchical representations.

What is a CNN?

A Convolutional Neural Network processes images by extracting spatial features:

- Local patterns (edges, corners, textures)
- Hierarchical structure (simple $\rightarrow$ complex)
- Parameter sharing across the image

Core layers:

- Convolution
- Activation (ReLU)
- Pooling
- Fully connected layers

How?

A convolution uses a small filter to detect patterns:

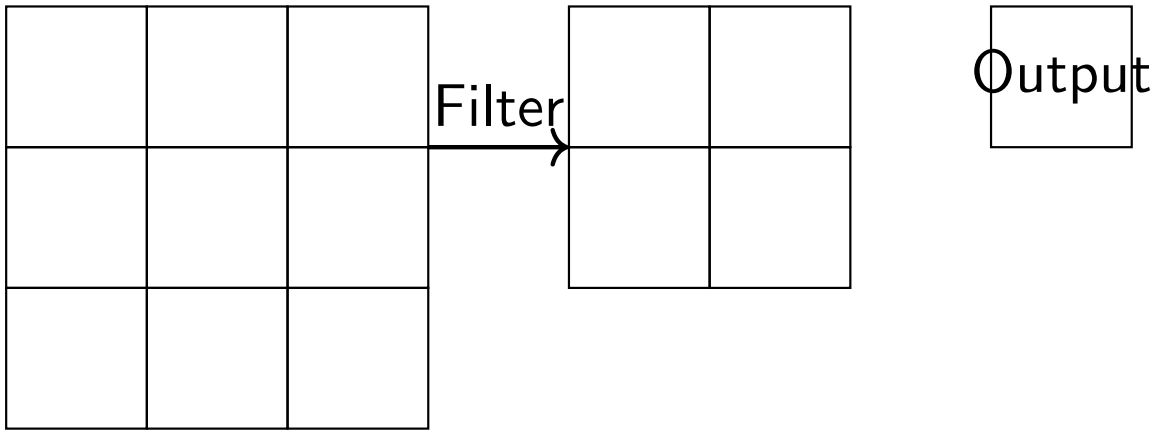
$$Y = X * W + b$$

- Filters slide across the image.
- Each filter learns a specific pattern.
- Produces a feature map.

Early filters:

- Detect edges, corners, gradients.

Diagram: Convolution



Numerical Example: Convolution

Image patch:

$$X = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}, \quad W = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$$

Compute:

$$Y = (2)(1) + (1)(-1) + (0)(2) + (3)(0)$$

$$Y = 2 - 1 = 1$$

Interpretation: Filter detects differences between top-left and top-right.

ReLU Activation

After convolution:

$$Z = X * W + b$$

Activate:

$$A = \max(0, Z)$$

Why ReLU?

- Removes negative values
- Introduces non-linearity
- Avoids vanishing gradient (better than sigmoid/tanh)

Why Padding?

Without padding, output shrinks:

$$\text{Output size} = n - k + 1$$

With SAME padding:

$$\text{Output size} = n$$

Reasons:

- Keeps spatial resolution
- Preserves edges
- Allows deeper CNNs

What is Stride?

Stride s = number of pixels the filter moves.

$$\text{Output size} = \frac{n-k}{s} + 1$$

Effects:

- $s = 1$: detailed scanning
- $s = 2$: downsampling

Advantages of Padding and Stride

Padding (P):

- Preserves spatial size.
- Ensures edge pixels contribute to the output.

Stride (S):

- Controls how far the filter moves per step.
- Larger stride $\rightarrow$ smaller output map.

Pooling: Downsampling Features

- Reduces spatial dimensions.
- Controls overfitting.

Max Pooling Example

$$\text{MaxPool}(2 \times 2) = \max\{4, 2, 1, 3\}$$

Pooling (Downsampling)

Max pooling example:

$$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \Rightarrow \max = 3$$

Benefits:

- Reduces dimensionality
- Keeps strongest features
- Provides translation invariance

Pooling Diagram

0	3
2	1

$\xrightarrow{\text{Max Pool}}$

3

Flattening

Transforms 3D tensor:

$$H \times W \times C$$

into 1D vector.
Purpose:

- Feed into Dense layers
- Combine high-level features

Fully Connected Layers

Standard neural network layer:

$$y = Wh + b$$

Role:

- Integrate extracted features
- Perform classification or regression

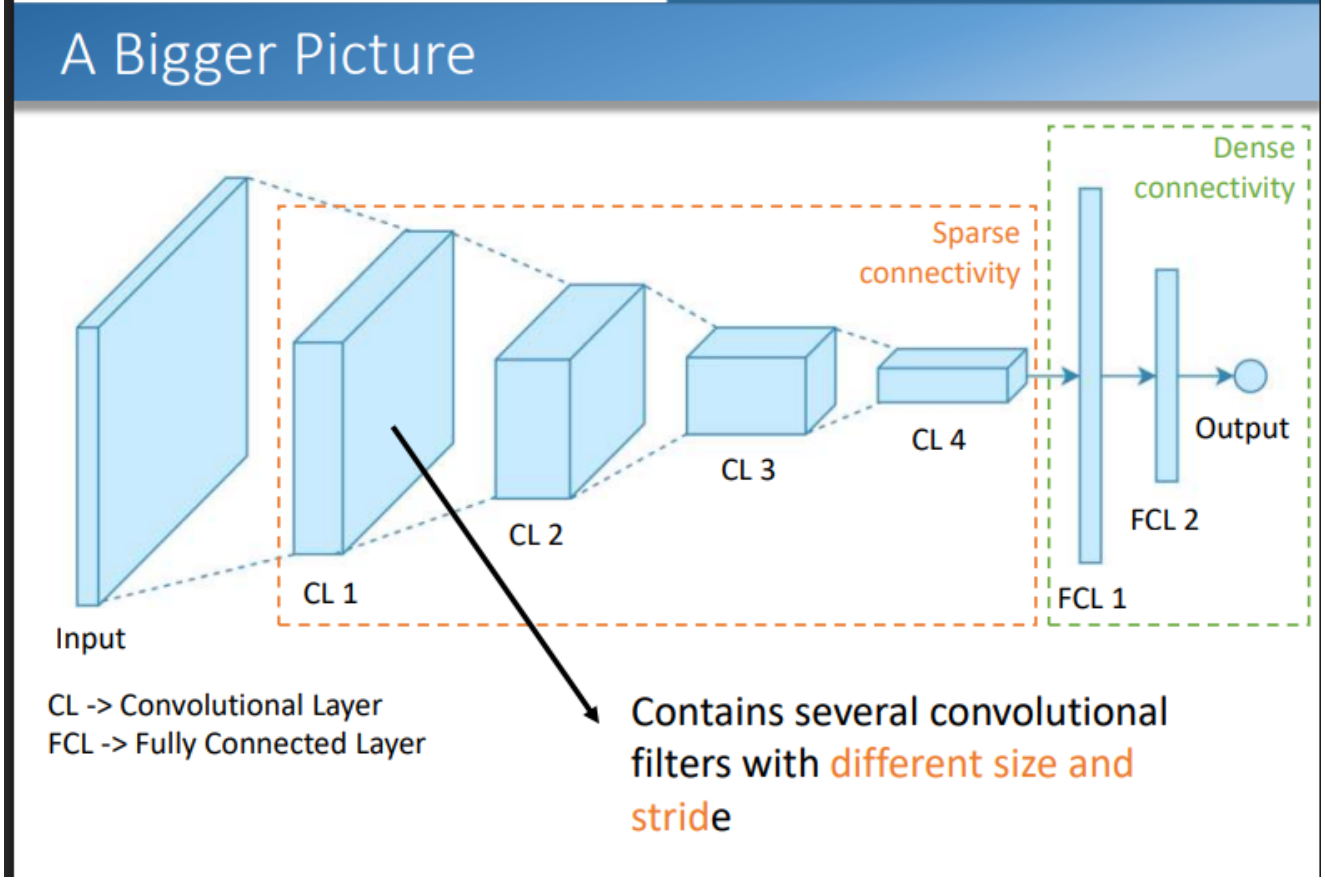


Figure: Bigger Picture

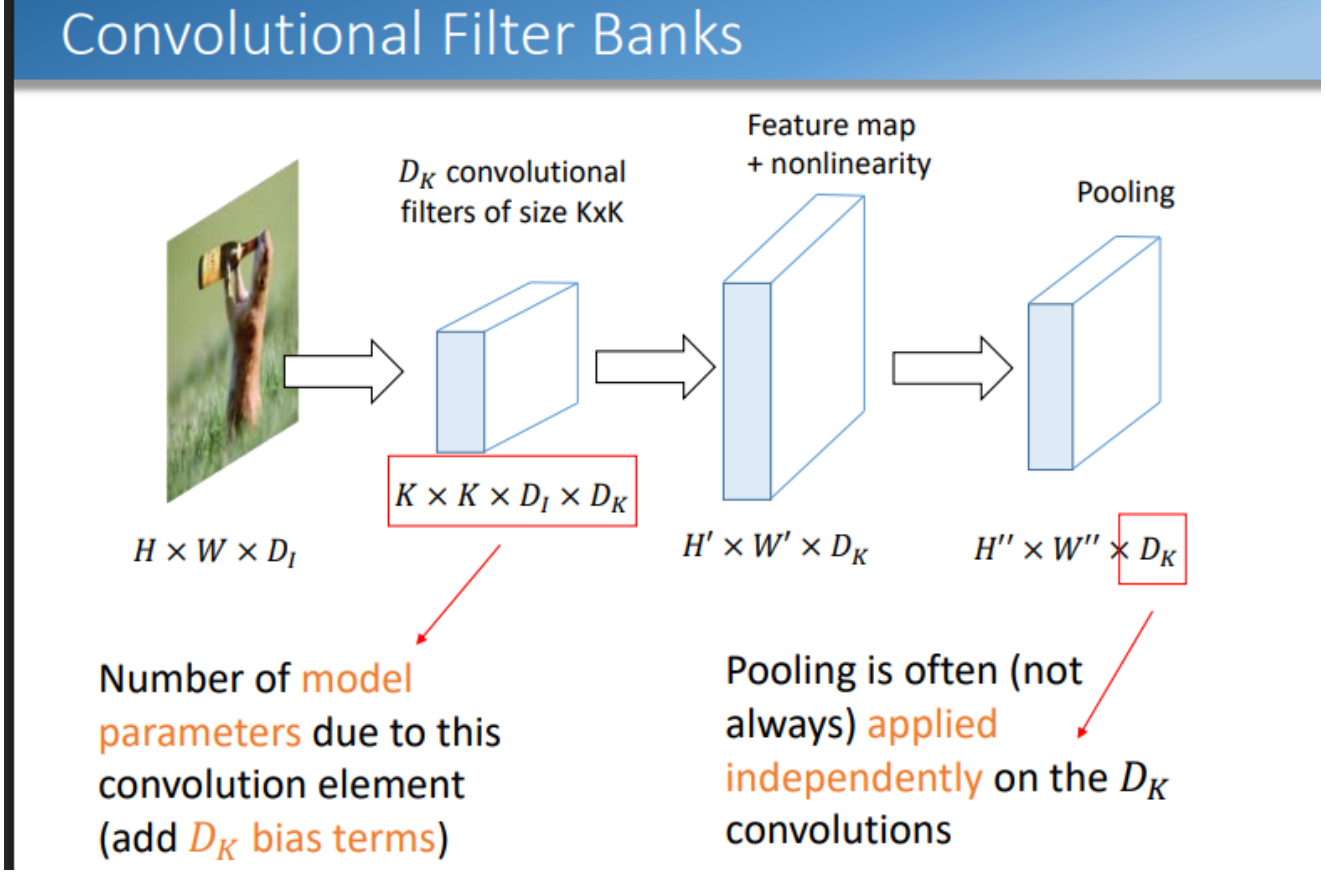


Figure: Convolutional Layers

CNN End-to-End Example

1. Input image: $32 \times 32 \times 3$
2. Conv layer: 32 filters $\rightarrow 32 \times 32 \times 32$
3. ReLU activation
4. Max pooling $\rightarrow 16 \times 16 \times 32$
5. Conv layer: 64 filters $\rightarrow 16 \times 16 \times 64$
6. Pooling $\rightarrow 8 \times 8 \times 64$
7. Flatten $\rightarrow$ vector of length 4096
8. Dense layer $\rightarrow$ 128 units
9. Softmax $\rightarrow$ 10 classes

Summary of CNN Steps

- Convolution extracts spatial patterns
- ReLU introduces non-linearity
- Padding preserves image size
- Stride controls downsampling
- Pooling reduces dimensionality
- Flatten converts features for Dense layers
- Fully connected layers classify

Specifying CNN in Code (Keras)

Number of convolution filters D_k

Define input size (only first hidden layer)

```
model = Sequential()
model.add(Conv2D(32, kernel_size=(5, 5), strides=(1, 1),
                 activation='relu',
                 input_shape=input_shape))
model.add(MaxPooling2D(pool_size=(2, 2), strides=(2, 2)))
model.add(Conv2D(64, (5, 5))
model.add(Activation='relu'))
model.add(MaxPooling2D(pool_size=(2, 2)))
model.add(Flatten())
model.add(Dense(1000, activation='relu'))
model.add(Dense(num_classes, activation='softmax'))
```

Does for you all the calculations to determine the final size to the dense layer (in most frameworks, you have to supply it)

Figure: Keras Example

Convolution: A Simple Example

Input (3x3):

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 1 & 1 \end{bmatrix}$$

Filter (2x2):

$$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Output (2x2):

$$\begin{bmatrix} (1*1+2*0+0*(-1)+1*1) & \cdots \\ \cdots & \cdots \end{bmatrix}$$

Interpretation: Filter highlights spatial structure in the input.

Multi-Step Convolution on a 4×4 Image

Input image:

$$X = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 3 & 1 & 2 & 2 \\ 0 & 1 & 3 & 1 \\ 2 & 0 & 1 & 2 \end{bmatrix}$$

Filter (kernel):

$$K = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Stride $= 1$, no padding. Output size:

$$(4 - 2 + 1) = 3 \Rightarrow 3 \times 3 \text{ output}$$

Step-by-Step Convolution (Top-Left Window)

Window:

$$\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$$

Compute:

$$(1)(1) + (2)(0) + (3)(-1) + (1)(1)$$
$$= 1 + 0 - 3 + 1 = -1$$

Place at $(1, 1)$ of output.

Next Window (Top-Middle)

Window:

$$\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$$

Compute:

$$(2)(1) + (0)(0) + (1)(-1) + (2)(1)$$
$$= 2 + 0 - 1 + 2 = 3$$

Place at $(1, 2)$.

Next Window (Top-Right)

Window:

$$\begin{bmatrix} 0 & 1 \\ 2 & 2 \end{bmatrix}$$

Compute:

$$(0)(1) + (1)(0) + (2)(-1) + (2)(1)$$
$$= 0 + 0 - 2 + 2 = 0$$

Place at $(1, 3)$.

Final Output Feature Map

After scanning all windows:

$$Y = \begin{bmatrix} -1 & 3 & 0 \\ 2 & 1 & 3 \\ -3 & 2 & 1 \end{bmatrix}$$

Interpretation:

- Positive responses indicate matched patterns.
- Negative responses indicate opposing patterns.

Input and Filter

Input Image (5×5):

$$X = \begin{bmatrix} 1 & 2 & 0 & 1 & 3 \\ 3 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 3 & 1 \\ 2 & 1 & 0 & 1 & 2 \\ 0 & 3 & 1 & 2 & 1 \end{bmatrix}$$

Filter (3×3):

$$K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

Stride $= 1$, Padding $= 0$. Output size:

$$(5 - 3 + 1) = 3 \quad \Rightarrow \quad 3 \times 3$$

Convolution Step: First Window

Window (top-left):

$$\begin{bmatrix} 1 & 2 & 0 \\ 3 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$$

Compute:

$$1(1) + 2(0) + 0(-1) + 3(1) + 1(1) + 2(0) + 1(0) + 0(-1) + 1(1) = 6$$

This becomes output element Y_{11}.

Complete Convolution Output

After computing all 9 windows:

$$Y = \begin{bmatrix} 6 & 6 & -1 \\ 1 & 1 & 7 \\ 1 & -1 & 0 \end{bmatrix}$$

Interpretation:

- Positive values $\rightarrow$ strong pattern match
- Negative values $\rightarrow$ opposite pattern

ReLU Activation

Apply ReLU:

$$R = \max(0, Y) = \begin{bmatrix} 6 & 6 & 0 \\ 1 & 1 & 7 \\ 1 & 0 & 0 \end{bmatrix}$$

Non-linear activation introduces deeper representational power.

Max Pooling (2×2, stride 1)

Pooling windows:

$$\begin{bmatrix} 6 & 6 \\ 1 & 1 \end{bmatrix} \Rightarrow 6 \qquad \begin{bmatrix} 6 & 0 \\ 1 & 7 \end{bmatrix} \Rightarrow 7$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow 1 \qquad \begin{bmatrix} 1 & 7 \\ 0 & 0 \end{bmatrix} \Rightarrow 7$$

Final pooled map:

$$P = \begin{bmatrix} 6 & 7 \\ 1 & 7 \end{bmatrix}$$

Flatten Layer

Flatten row-wise:

$$\text{Flatten}(P) = [6,\ 7,\ 1,\ 7]$$

This vector can now enter a fully connected layer.

VALID vs SAME Padding

VALID padding (no padding):

$$\text{Output size} = n - k + 1$$

Example (5×5 input, 3×3 filter):

$$5 - 3 + 1 = 3 \Rightarrow 3 \times 3$$

SAME padding: Pad so output size = input size:

$$p = \frac{k-1}{2} \Rightarrow p = 1$$

Add 1-pixel border around the image:

$$X_{padded} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 3 & 0 \\ 0 & 3 & 1 & 2 & 2 & 0 & 0 \\ 0 & 1 & 0 & 1 & 3 & 1 & 0 \\ 0 & 2 & 1 & 0 & 1 & 2 & 0 \\ 0 & 0 & 3 & 1 & 2 & 1 & 0 \end{bmatrix}$$

Full Worked Example Summary

1. **Input:** 5×5 image
2. **Conv (3×3 filter):** produces 3×3 feature map

$$Y = \begin{bmatrix} 6 & 6 & -1 \\ 1 & 1 & 7 \\ 1 & -1 & 0 \end{bmatrix}$$

3. **ReLU applied:**

$$R = \begin{bmatrix} 6 & 6 & 0 \\ 1 & 1 & 7 \\ 1 & 0 & 0 \end{bmatrix}$$

4. **Max Pooling (2×2, stride 1):**

$$P = \begin{bmatrix} 6 & 7 \\ 1 & 7 \end{bmatrix}$$

5. **Flatten:** $[6, 7, 1, 7]$
6. **Ready for Dense Layer + Softmax**

Backprop Through Convolution: Overview

Goal: Compute gradients of loss L w.r.t.

$$W, \quad b, \quad X$$

Forward:

$$Y_{ij} = \sum_{u,v} X_{i+u,j+v} \, W_{uv} + b$$

Backward:

$$\frac{\partial L}{\partial W_{uv}} = \sum_{i,j} \frac{\partial L}{\partial Y_{ij}} X_{i+u,j+v}$$

$$\frac{\partial L}{\partial X_{ij}} = \sum_{u,v} \frac{\partial L}{\partial Y_{(i-u)(j-v)}} W_{uv}$$

Derivative w.r.t. Filter Weights

Given:

$$Y = X * W$$

Using chain rule:

$$\frac{\partial L}{\partial W_{uv}} = \sum_{i,j} \frac{\partial L}{\partial Y_{ij}} \frac{\partial Y_{ij}}{\partial W_{uv}}$$

Since:

$$\frac{\partial Y_{ij}}{\partial W_{uv}} = X_{i+u,j+v}$$

Thus:

$$\boxed{\frac{\partial L}{\partial W_{uv}} = \sum_{i,j} \delta_{ij} X_{i+u,j+v}}$$

Where:

$$\delta_{ij} = \frac{\partial L}{\partial Y_{ij}}$$

Derivative w.r.t. Input Feature Map

Using:

$$Y_{ij} = \sum_{u,v} X_{i+u,j+v} W_{uv}$$

Then:

$$\frac{\partial L}{\partial X_{pq}} = \sum_{i,j} \delta_{ij} \frac{\partial Y_{ij}}{\partial X_{pq}}$$

Non-zero when $(p, q) = (i + u, j + v)$:

$$\boxed{\frac{\partial L}{\partial X_{pq}} = \sum_{u,v} \delta_{(p-u)(q-v)} \, W_{uv}}$$

This is effectively a full convolution of δ with flipped filter.

Numerical Backprop Example

Suppose:

$$Y = X * W$$

and the gradient from next layer is:

$$\delta = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$$

Let filter:

$$W = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

Compute gradient w.r.t W:

$$\frac{\partial L}{\partial W} = X \star \delta$$

Each weight's gradient is a sum of products of overlapping patches.
Example for W_{11}:

$$\frac{\partial L}{\partial W_{11}} = \sum_{i,j} \delta_{ij} X_{i,j}$$

Conceptual Overview of CNN Backpropagation

Backpropagation in CNNs is an elegantly structured mathematical process that enables the network to learn discriminative visual features from high-dimensional image data. Its power emerges from the interplay of:

- **Calculus**: Chain rule for computing gradients.
- **Linear Algebra**: Efficient tensor computations.
- **Numerical Optimization**: SGD, Momentum, Adam.

The mechanism updates filter weights through:

- gradient flow from deep layers $\rightarrow$ shallow layers,
- accumulation of gradients over all positions where a shared filter is applied,
- propagation through pooling layers and nonlinear activations.

This allows CNNs to excel in fields such as autonomous driving, medical imaging, security, and environmental sensing.

Mathematics of Convolution Backpropagation

For a filter K producing output feature map O, the gradient is:

$$\frac{\partial L}{\partial K(m,n)} = \sum_i \sum_j \frac{\partial L}{\partial O(i,j)} I(i+m, j+n)$$

Interpretation:

- Each filter weight receives a gradient from every position where it is applied.
- This is **weight sharing**: the same filter scans across the whole image.
- Backprop "slides" over the input just like the forward convolution.

Convolutional Layer Forward Pass

Given input I, filter F, and bias b:

$$O = (I * F) + b$$

where $*$ denotes convolution.

Activation:

$$O = \sigma\left((I * F) + b\right)$$

For ReLU:

$$\sigma(x) = \max(0, x)$$

Pooling: (max or average pooling)
Reduces spatial dimensions while preserving important features.

Loss Calculation

For classification, the cross-entropy loss is:

$$L = -\sum_{c} y_c \log(\hat{y}_c)$$

where $\hat{y}_c$ = predicted probability of class c $y_c = 1$ if correct class, else 0.

Loss quantifies the error between prediction and ground truth.

Backward Pass: Overview

Goal: Compute gradients $\frac{\partial L}{\partial \theta}$ for all parameters θ.
Main gradient components:

1. Gradient w.r.t. output of last layer
2. Gradient through activation function
3. Gradient w.r.t. convolution filters
4. Gradient w.r.t. bias
5. Gradient through pooling layers

All use the **chain rule**:

$$\frac{\partial L}{\partial A} = \frac{\partial L}{\partial B} \cdot \frac{\partial B}{\partial A}$$

Gradient w.r.t. Output

For the final layer output O:

$$\frac{\partial L}{\partial O}$$

This depends on the loss function. Example (cross-entropy + softmax):

$$\frac{\partial L}{\partial O} = \hat{y} - y$$

Gradient Through Activation Function

$$\frac{\partial L}{\partial I} = \frac{\partial L}{\partial O} \cdot \frac{\partial O}{\partial I}$$

For ReLU:

$$\frac{\partial O}{\partial I} = \begin{cases} 1 & I > 0 \\ 0 & I \leq 0 \end{cases}$$

Interpretation: Only activated neurons propagate gradients.

Gradient w.r.t. Convolution Filters

For convolution output:

$$O = I * F$$

The gradient w.r.t. filter weights:

$$\frac{\partial L}{\partial F} = \frac{\partial L}{\partial O} * \text{rot}_{180}(I)$$

where $\text{rot}_{180}(I)$ rotates the input by 180°.

Why rotate?

- Backprop uses correlation, not convolution.
- Rotation aligns indices to compute correct gradients.

To backpropagate into the input I:

$$\frac{\partial L}{\partial I} = \frac{\partial L}{\partial O} * \text{rot}_{180}(F)$$

Interpretation: Input receives gradients from all filter positions that used it.

Gradient w.r.t. Biases

Bias gradient is a simple sum:

$$\frac{\partial L}{\partial b} = \sum_{i,j} \frac{\partial L}{\partial O(i,j)}$$

Each output unit contributes to the bias.

Backprop Through Pooling Layers

Max Pooling:

- Gradient goes ONLY to the maximum value in each window.

Average Pooling:

$$\frac{\partial L}{\partial I_{ij}} = \frac{1}{k^2} \frac{\partial L}{\partial O}$$

Pool downsampling reduces spatial resolution of gradients.

Parameter Update Rule

Using Stochastic Gradient Descent:

$$F_{\text{new}} = F_{\text{old}} - \eta \frac{\partial L}{\partial F}$$

$$b_{\text{new}} = b_{\text{old}} - \eta \frac{\partial L}{\partial b}$$

- η = learning rate
- Similar updates apply to all parameters in CNN layers

Vanishing Gradients

In deep networks:

$$\left|\frac{\partial L}{\partial \theta}\right| \to 0$$

Causes:

- Sigmoid/tanh saturating regions
- Deep multiplication of small derivatives

Solutions:

- ReLU activation
- Batch normalization
- Residual connections

Exploding Gradients

Gradients become excessively large:

$$\left|\frac{\partial L}{\partial \theta}\right| \to \infty$$

Causes:

- Multiplication of large derivatives
- Unstable parameter initialization

Solutions:

- Gradient clipping
- Better weight initialization (He, Xavier)

Summary

Backpropagation in CNNs involves:

- Forward pass through convolution, activation, pooling
- Loss calculation
- Backprop through output, activation, convolution, pooling
- Updating filters and biases using SGD or Adam

Understanding the mathematics ensures correct model training and deeper insight into how CNNs learn complex visual patterns.

Practice Exercises- Q1: Manual Convolution Computation

Given the input:

$$I = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

with stride 1 and no padding, compute the output feature map.

Solution:

Patch (1,1):

$$1 \cdot 1 + 2 \cdot (-1) + 3 \cdot 0 + 1 \cdot 1 = 1 - 2 + 0 + 1 = 0$$

Patch (1,2):

$$2 \cdot 1 + 0 \cdot (-1) + 1 \cdot 0 + 1 \cdot 1 = 2 + 0 + 0 + 1 = 3$$

Patch (2,1):

$$3 \cdot 1 + 1 \cdot (-1) + 0 \cdot 0 + 2 \cdot 1 = 3 - 1 + 0 + 2 = 4$$

Patch (2,2):

$$1 \cdot 1 + 1 \cdot (-1) + 2 \cdot 0 + 2 \cdot 1 = 1 - 1 + 0 + 2 = 2$$

$$O = \begin{bmatrix} 0 & 3 \end{bmatrix}$$

Q2: Output Size With Stride and Padding

For an input of size 7×7, filter size 3×3, stride $s = 2$, padding $p = 1$:

$$\text{Output size} = \frac{N - F + 2p}{s} + 1$$

Solution:

$$\frac{7 - 3 + 2(1)}{2} + 1 = \frac{6}{2} + 1 = 3 + 1 = 4$$

So output is 4×4.

Q3: Sliding Window Interpretation

Explain in your own words what the sliding window (filter movement) accomplishes in a CNN.

Answer:

The sliding window allows filters to scan across the entire image, detecting patterns in every region. Instead of assigning a separate weight per pixel (like an MLP), CNNs reuse the same filter everywhere. This gives CNNs:

- translation invariance
- local pattern detection
- lower number of parameters

Q4: ReLU After Convolution

Given a convolution output:

$$O = \begin{bmatrix} -4 & 2 \\ 5 & -1 \end{bmatrix}$$

Apply ReLU activation.
Solution:

$$O_{\text{ReLU}} = \begin{bmatrix} 0 & 2 \\ 5 & 0 \end{bmatrix}$$

Q5: Parameter Count in a Conv Layer

How many learnable parameters are in a convolutional layer with:

- 32 filters
- filter size 3×3
- input depth = 3 (RGB)

Solution:
Each filter has:

$$3 \times 3 \times 3 = 27 \text{ weights}$$

plus a bias:

$$27 + 1 = 28$$

Total parameters:

$$32 \times 28 = 896$$

Much fewer than a fully connected layer.

Q6: CNN vs MLP on Images

Why does a CNN outperform an MLP on image classification?
Answer:

- CNNs use weight sharing $\rightarrow$ far fewer parameters.
- CNNs preserve spatial structure; MLPs flatten images.
- CNNs detect local features such as edges, textures, shapes.
- CNNs are translation invariant.

Q7: Pooling Operation

Given:

$$\begin{bmatrix} 4 & 1 \\ 7 & 3 \end{bmatrix} \quad \begin{bmatrix} 5 & 2 \\ 0 & 8 \end{bmatrix}$$

Apply 2×2 max pooling over each 2×2 block.

Solution:

$$\text{max pool} = \begin{bmatrix} 7 & 8 \end{bmatrix}$$

Q8: Compute Output Size (General Case)

Given:

$$N = 28, \quad F = 5, \quad p = 2, \quad s = 1$$

Solution:

$$\frac{28 - 5 + 2(2)}{1} + 1 = 28$$

Output remains 28×28. This is why padding is sometimes called **same padding**.

Q9: Stride 2 Example

Input width = 10, filter size = 3, stride = 2, padding = 0.

$$\frac{10 - 3 + 0}{2} + 1 = \frac{7}{2} + 1 = 3.5 + 1$$

Since output size must be integer:

$$\Rightarrow 4 \text{ (floor)}$$

Q10: Why Backprop Through CNN is Efficient

Explain briefly why backpropagation through a CNN is computationally efficient.

Answer:

- Filters are shared $\rightarrow$ gradients also shared $\rightarrow$ fewer parameters.
- Convolutions are structured $\rightarrow$ can reuse computations.
- GPUs accelerate convolution massively due to parallelism.

Q11: Gradient Descent Update Rule

Write the gradient descent update rule for a weight w with learning rate η.

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w}$$

Concept: Move weight opposite the direction of steepest ascent of the loss.

Q12: Learning Rate Too Large

What happens if the learning rate is too large?
Answer:

- The loss oscillates or diverges.
- The model fails to converge.
- Training becomes unstable.

Q13: Learning Rate Too Small

What happens if the learning rate is too small?
Answer:

- Very slow convergence.
- Model may get stuck in local minima or ravines.
- Training time becomes extremely long.

Q14: Multiple Filters Output Depth

If the input is $28 \times 28 \times 3$ and the convolution layer uses 64 filters, what is the depth of the output volume?

Answer: The depth equals the number of filters:

$$\text{depth} = 64$$

Q15: Flatten Layer Size

If you have a feature map of shape $16 \times 16 \times 32$, what is the length of the flattened vector?

$$16 \cdot 16 \cdot 32 = 8192$$

Answer: 8192.

Q16: Why CNNs Learn Hierarchical Features

Explain why deeper CNN layers learn more abstract patterns.
Answer:

- Early layers learn edges and simple textures.
- Middle layers learn shapes and object parts.
- Deep layers learn high-level concepts (faces, digits, etc.).

This hierarchical composition is natural for images.

Q17: Purpose of Padding

Why is padding used?
Answer:

- Prevent shrinking of output feature maps.
- Preserve spatial information near boundaries.
- Control output size.

Q18: Mini-Batch Gradient Descent in CNNs

Why do CNNs usually use mini-batch gradient descent?
Answer:

- More efficient than full batch.
- More stable than pure SGD.
- Allows parallelization on GPUs.

Q19: Numerical Backprop Example

Given:

$$w = 2, \quad \frac{\partial L}{\partial w} = 0.4, \quad \eta = 0.1$$

Compute the new weight:

$$w_{\text{new}} = 2 - 0.1(0.4) = 2 - 0.04 = 1.96$$

Answer: 1.96.

Q20: Interpretation of Feature Maps

Explain what a feature map represents.

Answer: A feature map stores the activation responses of a learned filter at different locations in the input. Bright regions mean the filter strongly detected its pattern (edge, curve, texture).

Key CNN Backprop Insights

- Gradients w.r.t filters are computed by convolving the input with the upstream gradient.
- Gradients w.r.t input are computed by convolving the gradient with the flipped filter.
- Weight sharing implies gradients accumulate across all spatial locations.
- This makes CNNs efficient and powerful for images.

Further Practice Exercises

1. Compute gradients for a 2×2 filter given a 3×3 input and loss gradients.
2. Explain weight sharing and how it reduces parameter count.
3. Derive gradients for max pooling for a given 2×2 region.
4. Modify the CNN to include BatchNorm and Dropout.
5. Visualize the learned filters of the first layer.

Summary

- CNNs extract spatial features using convolution + pooling.
- Backpropagation in CNNs involves structured gradient flow:
  - shared filter updates,
  - backprop through pooling,
  - backprop through activations.
- These mathematical principles enable CNNs to perform exceptionally well in vision tasks.

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